

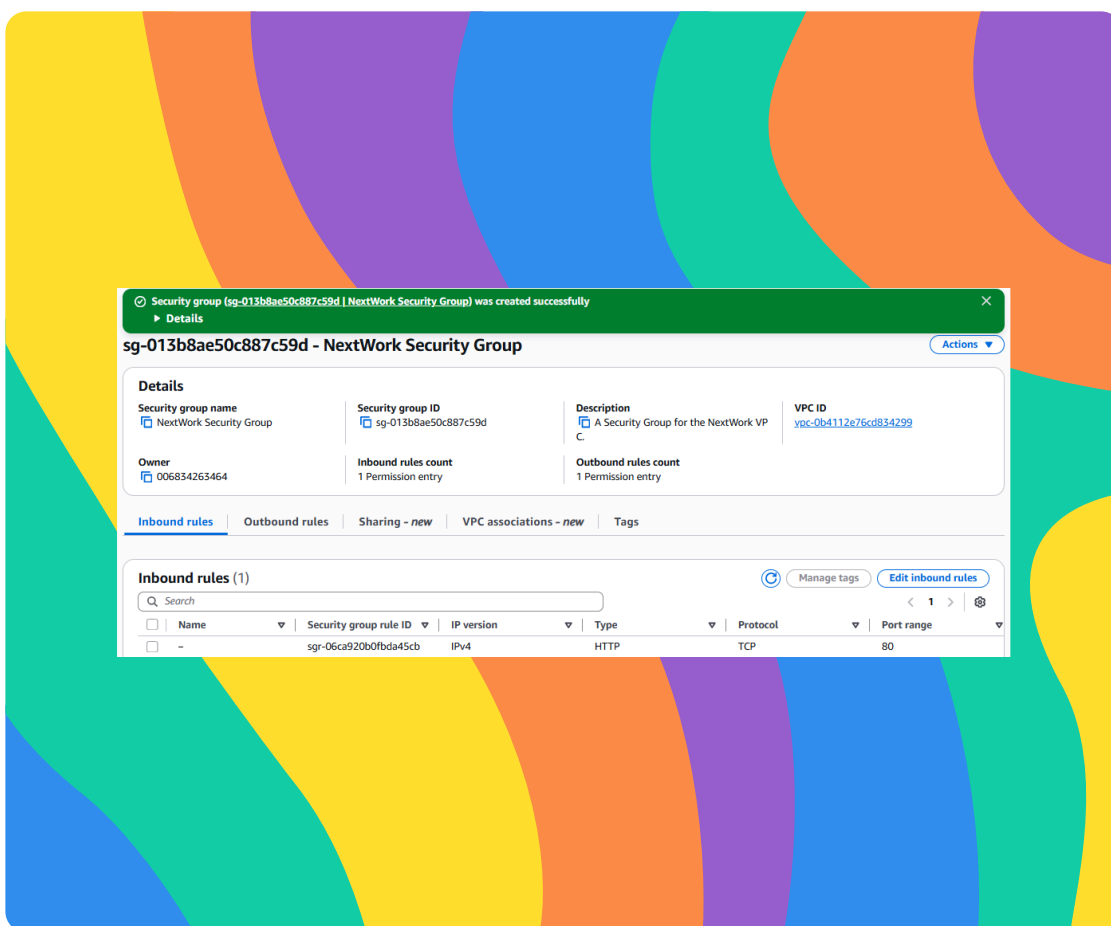


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VPC Traffic Flow and Security



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a private network in AWS, useful for building secure, flexible, and controlled cloud infrastructure.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a route table, security group, network ACL and track VPC resources across regions.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was to create VPC resources in another region and keep track of them using EC2 global view.

This project took me...

This project took me 2 hours to complete.



Route tables

Route tables are a data structure within a router or host that contains rules, or "routes," for directing network traffic to its destination. It functions like a map or timetable for network data, guiding packets along the most efficient path by specifying the "next hop" (the next router) and the network interface to use based on the packet's destination IP address.

Route tables are needed to make a subnet public because they provide a route to the Internet Gateway, enabling traffic between the subnet and the internet.

The screenshot displays the AWS Management Console interface for a Route Table. At the top, a green notification bar states: "Updated routes for rtb-0adbd48f52df0aa2d / NextWork route table successfully". Below this, the "Details" tab is active, showing the following information:

- Route table ID:** rtb-0adbd48f52df0aa2d
- VPC:** vpc-0b4112e76cd834299 | NextWork
- Main:** Yes
- Owner ID:** 006834263464
- Explicit subnet associations:** -
- Edge associations:** -

Below the details, the "Routes" tab is selected, showing a table of routes. The table has columns for Destination, Target, Status, Propagated, and Route Origin. There are 2 routes listed:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-025c195e707d12b	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table



Route destination and target

Routes are defined by their destination and target, which mean: Destination: the IP address range (CIDR block) that the route applies to. Target: the gateway or resource (e.g., Internet Gateway, NAT Gateway, local VPC) where traffic for that destination should be sent.

The route in your route table that directed internet-bound traffic to your internet gateway had a destination of 0.0.0.0/0 and a target of the Internet Gateway `igw-025c195e5e707d12b`

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Security groups

Security groups are virtual firewalls for your AWS resources that control inbound and outbound traffic at the instance level. They work by allowing (but never explicitly denying) traffic based on rules you define, such as protocol, port range, and source/destination IP or security group.

Inbound vs Outbound rules

Inbound rules are permissions that control what kind of incoming traffic is allowed to reach your resource (like an EC2 instance) through its security group. I configured an inbound rule that has a type HTTPS and source type Anywhere-IPv4.

Outbound rules are permissions that control what kind of outgoing traffic your resource is allowed to send out to other destinations. By default, my security group's outbound rule allows all outbound traffic (all protocols, all ports, to all destinations) so that instances can freely initiate connections to the internet or other resources.



Security group (sg-013b8ae50c887c59d | NextWork Security Group) was created successfully

Details

sg-013b8ae50c887c59d - NextWork Security Group

Details

Security group name
NextWork Security Group

Security group ID
sg-013b8ae50c887c59d

Description
A Security Group for the NextWork VP C

VPC ID
vpc-0b4112e76cd834299

Owner
006834263464

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

< 1 >

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-06ca920b0fbda45cb	IPv4	HTTP	TCP	80



Network ACLs

Network ACLs are a network-level layer of security, complementing security groups (which work at the instance level).

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security groups secure instances, while network ACLs secure subnets.

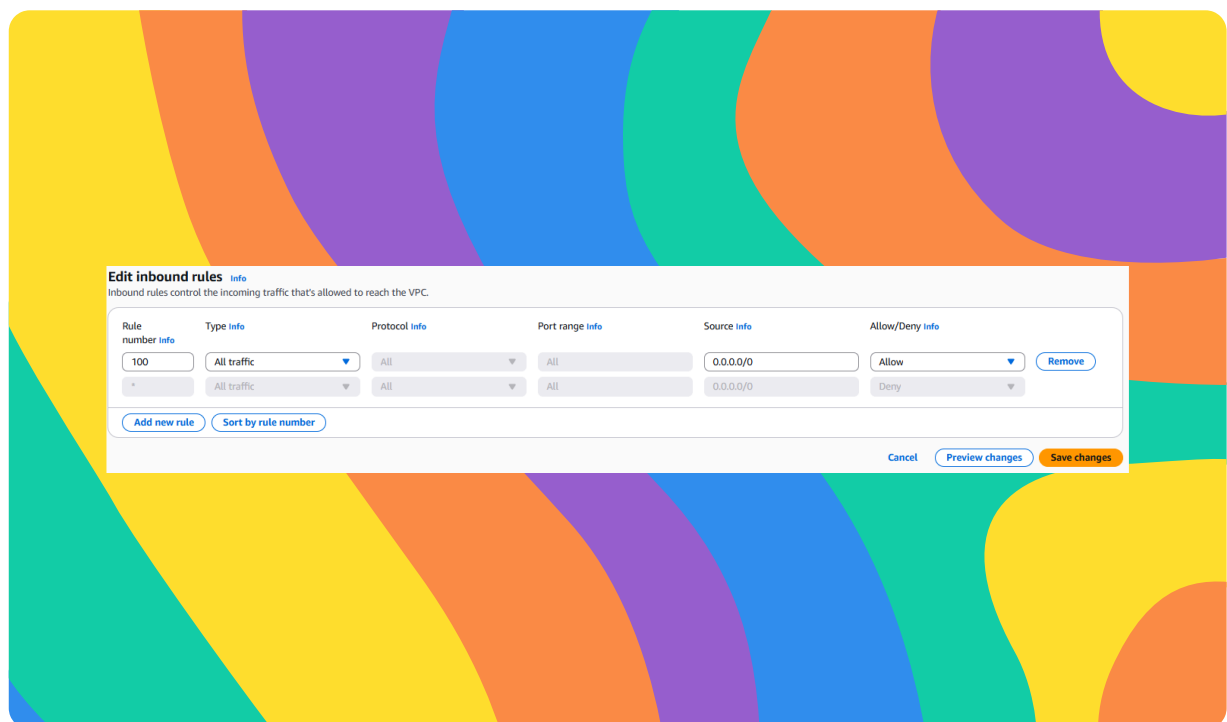


Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic in and out of the subnet until you modify them.

In contrast, a custom network ACL's inbound and outbound rules are automatically set to deny all traffic until you add rules to explicitly allow it.





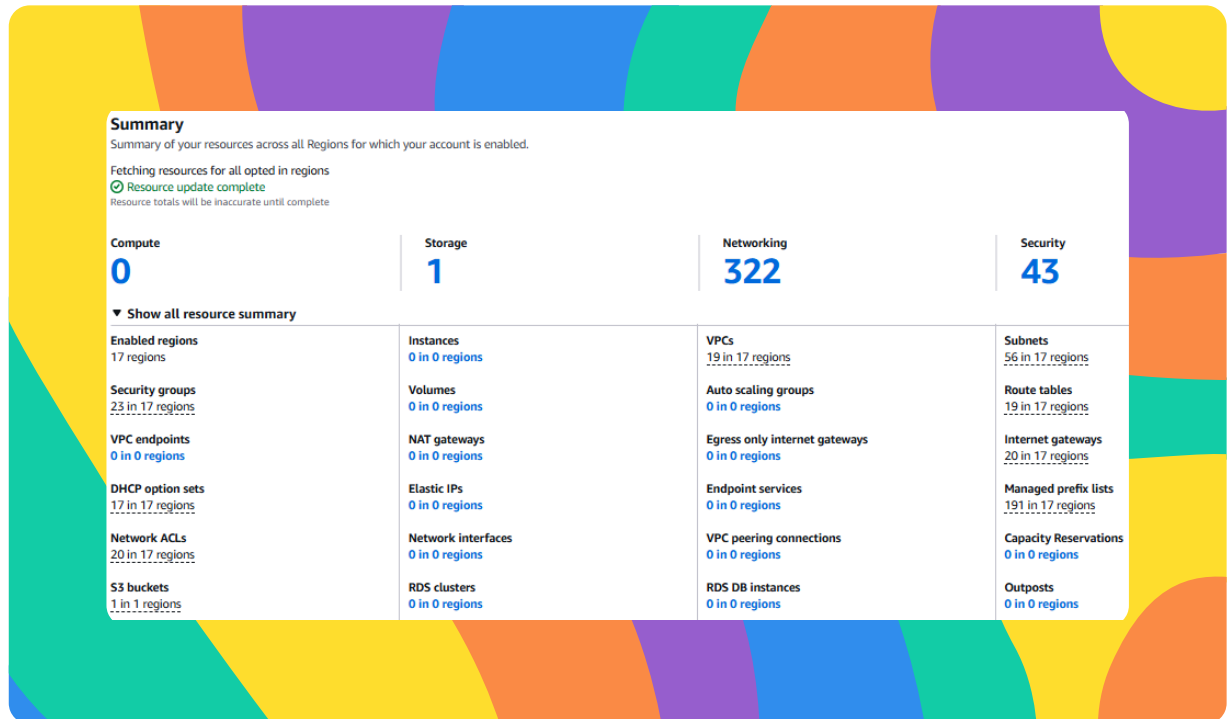
Tracking VPC Resources

I created additional VPC, internet gateway and security group. Instead of my usual region, I used us-west-1 which is North California. Teams would use multiple regions to:

1. Improve latency for their end users, which means apps run faster because resources are physically closer to the users' location.
2. Protect themselves from natural disasters and other risks - if one Region experiences an outage, other Regions can maintain operations.

EC2 Global View is a tool where you can find all your EC2 resources across every AWS Region in one centralized dashboard—including instances, VPCs, subnets, security groups, and more. I could even narrow down my search by filtering on specific Regions or specific resource types to quickly locate what I need. Without EC2 Global View, you'd have to manually switch between Regions in the AWS Management Console to track down and manage your resources, which is slower and more error-prone.

Now that I've learnt about EC2 Global View, I'd use it again to quickly monitor and manage my EC2 resources across multiple Regions in one place, instead of switching Regions manually. I'd also use it to troubleshoot, audit resources, or confirm where specific instances, VPCs, or security groups are located.





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